

* In the lecture we had a look at the HII regions of the ISM.

HI: Neutral Hydrogen (H)

HII: Ionized Hydrogen (H^+) + e^-

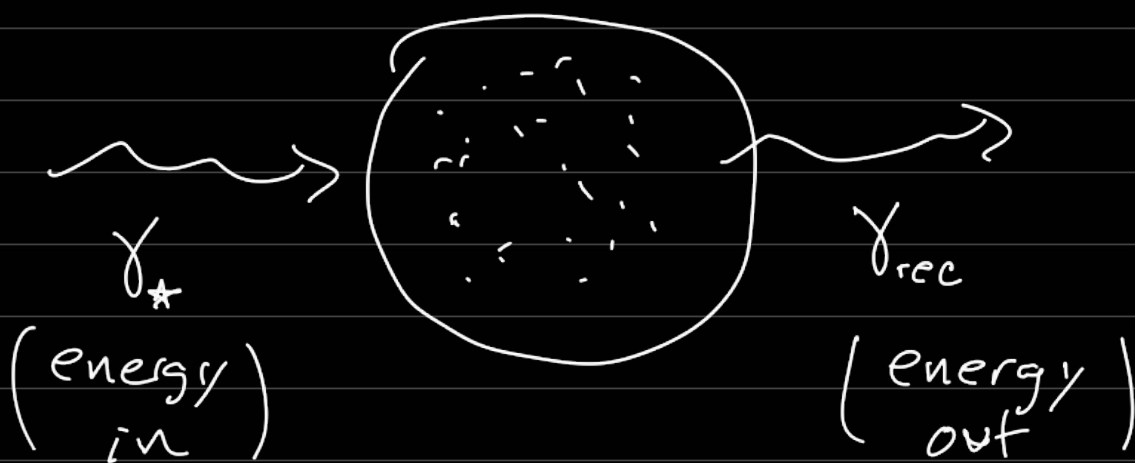
• Found HII region ionized by high energy photons



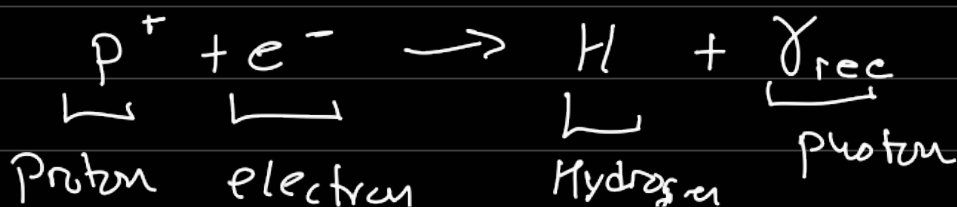
this is how the region is heated

Heating: Energy coming into the system

Cooling: Energy leaving the system



* Cooling: looked at recombination as a cooling mechanism



γ_{rec} has energy

$$\langle E_{\text{rec}} \rangle = k_B T_e$$

temp of electron

* We found an expression for the outward radius $R_I(t)$ of the ionization front via (see session notes)

$$4\pi R_I^2 n_H(R_I) \Delta R_I = S_* \Delta t$$



Then found the total recombinations

$$R_{\text{tot}} = N_e \propto n_p$$

$$N_e = n_e V \quad \leftarrow \text{Volume}$$

$$R_{\text{tot}} = n_e V \propto n_p$$

- To find the radius where equilibrium occurs we made the equality:

$$\underbrace{R_{\text{tot}}}_{\text{total}} = \underbrace{S_*}_{\text{total ionizing}}$$

(recombinations per sec) (photons per sec)

for PS1, (2), (b) we look at the differential:

$$\Delta R_{\text{tot}} = n_e \Delta V \alpha n_p$$

we saw

$$\Delta V = 4\pi R^2 \Delta R$$

$$\Delta R_{\text{tot}} = n_e \alpha n_p 4\pi R^2 \Delta R$$

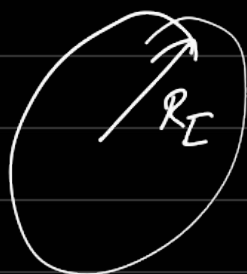
$\Delta R \rightarrow dR$, integrate

$$\int dR_{\text{tot}} = R_{\text{tot}} = \int n_e \alpha n_p 4\pi R^2 dR$$

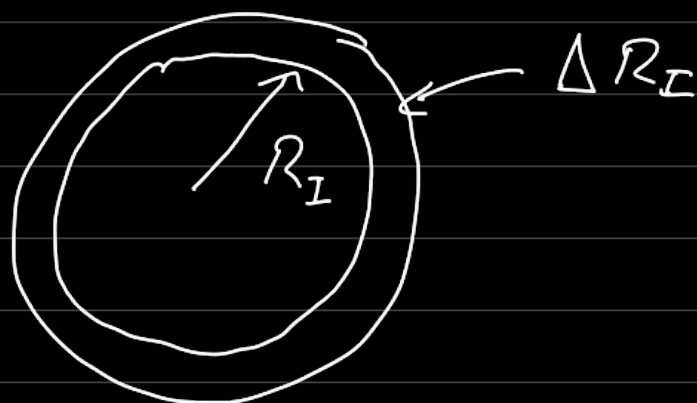
and assume

$$n_e = n_p = n_H$$

* Note the Strömberg radius R_S that we found is only at a given radius R_S . We can derive an expression for the general radius of the outward moving R_I



Why is this different from R_I we found earlier? Well, we only assumed R was dependent on ionization. Here, we must account for recombination:



• Recall we found

$$N_\gamma = S_* \Delta t = \left(\text{Total \# of ionizing photons} \right)$$

but $n_H = \frac{N_\gamma}{\Delta V}$

$$n_H \Delta V = S_* \Delta t$$

$$\boxed{n_H (4\pi R_I^2 \Delta R) = S_* \Delta t}$$



this is the expression without considering recombination

we know that

$$(\textcolor{red}{R}_{\text{tot}} \Delta t) = \left(\begin{array}{l} \text{total \# of recombs in} \\ \text{a time } \Delta t \end{array} \right)$$

this term must added to the above equation but it must be negative since this is the cooling which causes us to lose energy.

$$\textcolor{violet}{R}_{\text{tot}} \Delta t = \alpha n_e n_p V \Delta t = \alpha n_e n_p \left(\frac{4}{3} \pi R_I^3 \right) \Delta t$$

→ Add - this term to ~~⊗~~

$$\boxed{n_H 4\pi R_I^2 \Delta R = S_* \Delta t - \alpha n_p n_e \frac{4}{3} \pi R_I^3 \Delta t}$$

Heating
+

Cooling
-

* This is the equation used for PS1
(21) - (c)

Remember: $n_e = n_p = n_H$

Notes page 10 equation (28)

